# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET A

Chapters 11 & 12: **Keeping Time with the Skies, How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** reflects sunlight | Non-luminous |
| 2 | **(c)** waxing phase | After amavasya |
| 3 | **(c)** Moon's phases | Months follow phases |
| 4 | **(a)** day | About 24 hours |
| 5 | **(d)** solar calendar | Based on tropical year |
| 6 | **(a)** abiotic components | Air, water, soil, temperature |
| 7 | **(a)** aquatic ecosystem | Water-based |
| 8 | **(c)** carnivores | Flesh-eaters |
| 9 | **(c)** recycling nutrients | Nutrients return to soil |
| 10 | **(d)** frog | Eats a herbivore |

### SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Time between successive spring equinoxes (1). Basis of the Gregorian calendar (1). | 1+1 |
| 12 | Time for the same stars to rise at sunset again (1). Slightly longer than tropical year (1). | 1+1 |
| 13 | One benefits, the other is unaffected (1). Orchid on a tree (1). | 1+1 |
| 14 | One organism benefits and the other is harmed (1). Tapeworm in human gut (1). | 1+1 |

### SECTION — C (Short Answer) — Answers 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Collides with working satellites (1). Larger pieces can fall on Earth (1). Countries work together to remove it (1). | 1+1+1 |
| 16 | Deforestation (1). Pollution (1). Invasive species or overexploitation (1). | 1+1+1 |
| 17 | Researcher in space science and nuclear physics (1). Father of Indian Space programme (1). VSSC is named after him (1). | 1+1+1 |
| | **OR:** Go out after sunset or before sunrise (1). Look for a steadily moving point of light (1). Use apps for pass time (1). | 1+1+1 |
| 18 | Protect habitats (1½). Conserve biodiversity (1½). | 1½+1½ |
| | **OR:** Bandipur – Karnataka (1). Manas – Assam (1). Chilika – Odisha (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Debris from old satellites and rockets (1). Collision risk (1). Small pieces burn up (1). Larger pieces may crash (1). International cooperation to remove (1). | 1+1+1+1+1 |
| | **OR:** Sarabhai's fields (1). Father of Indian Space programme (1). First satellites effort (1). VSSC (1). Benefits to people (1). | 1+1+1+1+1 |
| 20 | Deforestation (1). Pollution (1). Climate change and invasive species (1). Plant trees (1). Reduce, reuse, recycle (1). | 1+1+1+1+1 |
| | **OR:** National parks (1). Sanctuaries (1). Biosphere reserves (1). Community areas (1). Importance (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET B

Chapters 11 & 12: **Keeping Time with the Skies, How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Earth takes slightly less than 365¼ days | Fine correction |
| 2 | **(c)** 22 March (21 in leap years) | The day after spring equinox |
| 3 | **(a)** lunar year is shorter than the solar year | 354 days |
| 4 | **(b)** AstroSat | ISRO satellite |
| 5 | **(c)** Thiruvananthapuram | Rocket and launch vehicle technology |
| 6 | **(c)** decrease | Less food available |
| 7 | **(c)** parasitism | Parasite benefits, host harmed |
| 8 | **(c)** organic manure | Vrikshayurveda advocates it |
| 9 | **(a)** 1987 | As per the chapter |
| 10 | **(a)** Karnataka | Where predators and prey were studied |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Equinox: Sun rises exactly east, equal day and night (1). Solstice: Sun reaches extreme positions (1). | 1+1 |
| 12 | Mapping and weather observation (1). Communication and space research (1). | 1+1 |
| 13 | Clean air and water (1). Food, medicine or climate regulation (1). | 1+1 |
| 14 | Deforestation (1). Pollution or overexploitation (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Rotation on axis defines a day (1½). Revolution round the Sun defines a year (1½). | 1½+1½ |
| 16 | Provide conditions to live (1½). Examples: sunlight, water, soil, temperature (1½). | 1½+1½ |
| 17 | Note its position and shape each day (1). Position shifts eastward (1). Shape of the bright part changes (1). | 1+1+1 |
| | **OR:** Months were adjusted to add up to 365 days (1½). February has 28 or 29 days (1½). | 1½+1½ |
| 18 | Natural: forest, pond (1½). Human-made: farm, fish pond (1½). | 1½+1½ |
| | **OR:** Energy is lost at each level (1½). Less energy supports fewer organisms (1½). | 1½+1½ |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Difference = 11¼ days (2). Accumulates over years (1). Extra month added about every three years (2). | 2+1+2 |
| | **OR:** Same time every evening (1). Draw the shape (1). Note position and rise time (1). Plot a chart (1). Identify pattern and cycle (1). | 1+1+1+1+1 |
| 20 | (i) Algae → small fish → big fish (1). (ii) Producer algae, consumers fish (1). (iii) Decomposers recycle (1). (iv) Big fish suffer (1). Algae increase (1). | 1+1+1+1+1 |
| | **OR:** Water decreases (1). Plants dry (1). Herbivores starve (1). Carnivores decline (1). Decomposers also affected (1). | 1+1+1+1+1 |

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## ANSWER KEY & MARKING SCHEME SET C

Chapters 11 & 12: **Keeping Time with the Skies, How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** we see different parts of its lit half as it orbits Earth | Relative positions |
| 2 | **(c)** decreases | Waning |
| 3 | **(a)** seasons (Earth's revolution around Sun) | 365 days |
| 4 | **(c)** 354 days | 12 × 29.5 |
| 5 | **(b)** 29 days | Extra day every four years |
| 6 | **(d)** soil | Non-living |
| 7 | **(d)** terrestrial ecosystem | Land-based |
| 8 | **(c)** omnivores | e.g. humans, crows |
| 9 | **(c)** food chain | Linear feeding order |
| 10 | **(c)** eagle | Fourth trophic level |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Old satellites and rocket parts in orbit (1). Can collide with working satellites (1). | 1+1 |
| 12 | Space scientist, Father of the Indian Space programme (1). Pioneered the first artificial satellites effort (1). | 1+1 |
| 13 | Non-native species that spread widely (1). They harm native species (1). | 1+1 |
| 14 | Growing the same crop repeatedly (1). Reduces biodiversity and affects pollinators (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Planning agriculture (1). Fixing festivals (1). Keeping records of events (1). | 1+1+1 |
| 16 | Dead matter would pile up (1). Nutrients not recycled (1). Plants and ecosystem would suffer (1). | 1+1+1 |
| 17 | Earth revolves around the Sun (1½). Seasons repeat as Earth moves between equinoxes (1½). | 1½+1½ |
| | **OR:** Lunar year = 12 × 29.5 = 354 days (1). Gap = 11 days (1). About 3 years give ~33 days, one extra month (1). | 1+1+1 |
| 18 | They make food using sunlight (1½). All consumers depend directly or indirectly on them (1½). | 1½+1½ |
| | **OR:** Clean air (1). Fresh water (1). Food and medicine (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Moon cycle sets a month (1). Earth's rotation sets a day (1). Earth's revolution sets a year (1). Seasons depend on position in orbit (1). Calendars combine these cycles (1). | 1+1+1+1+1 |
| | **OR:** No, Moon is always half lit (1). Phase depends on our view (1). Sun, Earth, Moon positions (1). Sketch with arrows (1). Cycle repeats (1). | 1+1+1+1+1 |
| 20 | Clean air and water (1). Food and medicine (1). Climate regulation (1). Human survival depends (1). Conservation protects future (1). | 1+1+1+1+1 |
| | **OR:** Vrikshayurveda text (1). Organic manure (1). Kunapa Jala (1). Chemical farming harms soil (1). Balanced approach (1). | 1+1+1+1+1 |

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## ANSWER KEY & MARKING SCHEME SET D

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### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** tropical year | Basis of Gregorian calendar |
| 2 | **(a)** Chaitra | Starts after the spring equinox |
| 3 | **(c)** equinoxes | Around 21 March and 23 September |
| 4 | **(b)** mapping and Earth observation | Remote sensing |
| 5 | **(b)** ancient Indian astronomical text | Mentions Makar and solstices |
| 6 | **(a)** both organisms benefit | e.g. flower and bee |
| 7 | **(d)** clean air, water, food and climate regulation | Benefits from nature |
| 8 | **(c)** liquid fertiliser from animal and plant waste | Made by fermentation |
| 9 | **(c)** deforestation | Habitat loss |
| 10 | **(b)** producer → herbivore → carnivore | Eaten by next level |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Makar Sankranti (1). Pongal or Bihu (1). | 1+1 |
| 12 | Moon moves along its orbit around Earth (1). It rises about 50 minutes later (1). | 1+1 |
| 13 | They slow winds and waves in storms (1). They are home to many species (1). | 1+1 |
| 14 | National parks (1). Wildlife sanctuaries or biosphere reserves (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | No bright part visible (1½). Thin crescent seen just after new moon in the evening (1½). | 1½+1½ |
| 16 | Tigers eat deer and wild boar (1). If prey declines, tigers starve (1). Needs healthy forests (1). | 1+1+1 |
| 17 | Stars in the background of the Sun marked the seasons (1½). This helped create accurate calendars (1½). | 1½+1½ |
| | **OR:** Take images of clouds and cyclones (1½). Give early warnings (1½). | 1½+1½ |
| 18 | They depend on the ecosystem (1½). Their knowledge helps manage it (1½). | 1½+1½ |
| | **OR:** Plant trees (1). Reduce pollution and plastic (1). Save water and energy (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Weather forecasts (1). Cyclone warnings (1). Communication and TV (1). Mapping by Cartosat (1). Agricultural planning (1). | 1+1+1+1+1 |
| | **OR:** Type (1). Start of year (1). Months (1). Leap year rule (1). Use (1). | 1+1+1+1+1 |
| 20 | Terrestrial examples (1). Aquatic examples (1). Abiotic differences (1). Organisms (1). Dependence between them (1). | 1+1+1+1+1 |
| | **OR:** Web diagram with arrows (1). Producer identified (1). Consumers levels named (1). Removal increases mice and frogs (1). Hawks lose a food source and balance changes (1). | 1+1+1+1+1 |

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